

Introduction to Artificial Intelligence

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, understanding language, and recognizing images.

2. What is Machine Learning?

Machine Learning (ML) is a subset of Artificial Intelligence. It allows computers to learn patterns from data instead of being explicitly programmed for every task. Common types of machine learning include supervised learning, unsupervised learning, and reinforcement learning.

3. What is Generative AI?

Generative AI refers to systems that can create new content such as text, images, audio, video, or computer code. Large Language Models (LLMs) are a type of generative AI system designed to understand and generate human language.

4. What is Retrieval-Augmented Generation (RAG)?

Retrieval-Augmented Generation, commonly called RAG, combines information retrieval with text generation. When a user asks a question, the application searches a knowledge source for relevant information. The retrieved content is then provided to a language model so that it can generate a more relevant answer based on the available documents.

5. Main Components of a RAG Application

Document Loader: Reads documents such as PDFs, text files, or web pages.

Text Splitter: Divides large documents into smaller sections called chunks.

Embedding Model: Converts text into numerical vectors that represent its meaning.

Vector Database: Stores embeddings and helps retrieve relevant chunks.

Retriever: Finds the document chunks most related to the user's question.

Language Model: Uses the question and retrieved context to generate an answer.

6. Example RAG Workflow

A user asks: 'What is RAG?' The system converts the question into an embedding, searches the vector database, retrieves relevant sections from the document, and sends those sections with the question to the language model. The model then produces an answer using the retrieved context.

7. Benefits of RAG

RAG can help applications answer questions using private or specialized documents. It can reduce the need to retrain a model whenever documents change and can make answers more traceable by connecting responses to retrieved source content.

8. Sample Questions for Testing

- What is Artificial Intelligence?
- Explain machine learning in simple words.
- What is Generative AI?
- What does RAG stand for?
- List the main components of a RAG application.
- Explain the RAG workflow.